

Container Orchestration with Kubernetes

Kubernetes is a production-grade open source container orchestration tool. It automates the deployment, scaling and management of containerised applications. Along with Kubernetes, this article will help you get acquainted with its package manager, Helm—its charts will help you install and even upgrade complex Kubernetes applications.



Did you know that with the appropriate settings, you can run more than 1000 lightweight virtual machines on your desktop computer? You don't need high-end servers to run such a large number of containers. Also, managing all the containers manually is a cumbersome task. To ease this task, we can use open source tools like Docker Swarm, Mesos, CoreOS Fleet, etc.

An introduction to Kubernetes

The use of containers increased tremendously with the advent of the Docker tool. For a large scale single application such as an online shopping system, multiple containers are used to achieve high availability, continuous deployment, failover mechanisms, high performance, etc. In such cases, the number of containers deployed is huge, and depends on the requirements. Sometimes the containers are deployed on multiple hosts. Since all these

containers and hosts needed to be managed, the concept of 'orchestration of containers' came about.

Kubernetes is the latest buzz word in the world of orchestrating containers. It is an open source container orchestration tool with a great set of features. Kubernetes is widely used amongst other container orchestration tools.

This article guides readers on how to set up a Kubernetes cluster using Minikube in an Ubuntu system. There are many ways to install a Kubernetes cluster, depending on the infrastructure being used. Using Minikube is one of the straightforward methods. Minikube abstracts lots of internals such as initialising the core pods, pod networking and exposing services, and makes it easy to fire up a Kubernetes cluster.

So let us set up and run a Kubernetes cluster locally with Minikube and then explore how to deploy applications on to

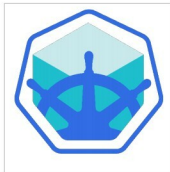


Figure 1: Minikube logo (Source: <https://github.com/kubernetes/minikube/>)